

2. Circles and Dots (20 marks)

Problem designed by Anindya Sharma and Monojit Choudhury

Inspired by Sherlock Holmes in “The Adventure of the Dancing Men”, Golu and his best friend Hammie decided to invent a code language for writing letters to each other. How cool, they thought, it would be to have a language that only they understood! However, they soon realized that drawing the 26 dancing men figures, a different one for each letter of the English alphabet, was not their cup of tea. Golu then had this brilliant idea that they could instead use fewer and simpler symbols: a dot (•), a circle (○) and a circle with a dot (⊙), and each English letter could be defined by a unique sequence of these symbols. He came up with the following scheme for encoding the first 7 letters of the alphabet:

Scheme 1 – A: •○○⊙, B: •⊙⊙, C: ••○, D: ○○○, E: ○•, F: ○○, G: ⊙

In this scheme, they would encode the word CAB as: ••○•○○⊙•○○

Hammie’s elder brother Shan, who was studying computer science, happened to pass by. He saw Golu’s code and immediately pointed out that this coding *scheme* won’t work. They would be in trouble, though he left it for Golu and Hammie to figure out why it won’t work and what scheme would work. Not sure of the solution, Golu and Hammie thought of designing 4 more coding schemes:

Scheme 2 – A: ○⊙, B: ⊙•, C: ⊙⊙, D: ○○○, E: ⊙•⊙, F: ○○○•, G: ○○○

Scheme 3 – A: •, B: ○, C: ⊙, D: ••, E: •○, F: •⊙, G: ○•

Scheme 4 – A: ○•, B: ○○, C: ⊙⊙, D: ••, E: ⊙○, F: ⊙•, G: ○⊙

Scheme 5 – A: ⊙•○, B: ⊙•⊙, C: ⊙○○, D: ⊙•○○, E: ⊙••, F: ⊙○○, G: ○○

They went to Shan, who now said that schemes 2 and 4 were fine; but schemes 3 and 5 have the same problem as scheme 1, and should not be used. Golu looked at the 5 schemes carefully and exclaimed: “Eureka! I know what the problem is with schemes 1, 3 and 5. I feel so stupid to have come up with such schemes!”

Shan was not sure though if Golu really understood the problem. He showed them 3 more schemes and asked them to figure out which of these would work.

Scheme 6 – A: ••, B: •○, C: •○○, D: ○•, E: ○○, F: ○⊙, G: ••○

Scheme 7 – A: ⊙⊙, B: ⊙○, C: ⊙•, D: ○•, E: ○○, F: ○⊙, G: •

Scheme 8 – A: •○○, B: •○○○○, C: ○⊙○○○, D: ○⊙⊙•, E: ○••••, F: ○⊙○•, G: •○•○

Assignment 1: What do you think is the problem with the schemes 1, 3 and 5, that schemes 2 and 4 do not have?

Assignment 2 [Answer Booklet]: Among the schemes 6, 7 and 8, which one(s) has/have the same problem as 1, 3 and 5?

Golu had indeed got his answers right. But now that Hammie and he had to extend their scheme for the remaining 19 letters, they needed a systematic process for designing coding schemes which would avoid the problem found in schemes 1, 3 and 5. Shan gave them a hint. He said “of the first 5 schemes, scheme 4 is special because none of the codes is a prefix of the code for another letter.” Such codes are called *prefix codes*. Scheme 3, for example, is not a prefix code because the code for A is a prefix of the code for D and E; similarly the code for B is a prefix of the code G.

Assignment 3 [Answer Booklet]: Is it true that any prefix code will avoid the problem that schemes 1, 3 and 5 have? Explain why or why not.

Assignment 4 [Answer Booklet]: Is it true that for a code to be useful, it must be a prefix code? Explain why or why not.

Notes: This problem is based on *Coding theory* which is the study of the properties of coding schemes and their fitness for a specific application. Codes are used for data compression, cryptography, error-correction, and networking. Codes are studied by various scientific disciplines—such as information theory, electrical engineering, mathematics, linguistics, and computer science. In 1948, scientist Claude Shannon published an article - "A Mathematical Theory of Communication" which laid the foundations of coding theory. Marcel J. E. Golay of *Golay code* fame and Richard Hamming who lent his name to *Hamming codes*, *Hamming windows*, *Hamming numbers*, and *Hamming distance* were some of the pioneers of this field.
